Section 10: The Nervous System & Motor Control

The nervous system is the body's communication and control network. For coaches, understanding the nervous system explains how movement is controlled, why technique improves with practice, how fatigue limits performance, and why recovery between maximal sessions requires more than just muscle repair.

10.1 Central and Peripheral Nervous Systems

- **Central Nervous System (CNS)** — the brain and spinal cord. The brain processes information, plans movements, and sends commands. The spinal cord mediates spinal reflexes without involving the brain.
- **Peripheral Nervous System (PNS)** — all nerves outside the CNS. Includes sensory neurons and motor neurons.

10.2 Proprioception — The Body's Position Sense

Proprioception is the body's ability to sense its own position and movement in space without visual input, mediated by receptors in muscles, tendons, and joints.

Proprioception is critical for efficient climbing footwork. Experienced climbers place their feet precisely without looking, because their proprioceptive system has been trained through thousands of repetitions. Beginners must look at every foothold since their proprioceptive maps have not yet developed.

This is why technical volume — time on easy terrain with deliberate focus on movement quality — is irreplaceable for developing climbing skill. Proprioception requires repetitive, varied movement practice on the wall, not gym exercises alone.

10.3 Neural Fatigue

Neural fatigue is distinct from muscular fatigue. After maximal effort training, the CNS requires recovery time separate from muscle recovery. Signs of accumulated neural fatigue include:

- Performance decline despite feeling physically fresh
- Reduced motivation and mental sharpness
- Slower reaction times and coordination errors
- Difficulty recruiting muscle to full capacity

Full CNS recovery from a single maximal effort session takes approximately 48–72 hours. Consecutive high-intensity days without this window leads to compounding neural fatigue and elevated injury risk — a strong argument for structured rest days.

10.4 Motor Learning — How Technique Improves

Motor learning is how the nervous system acquires, refines, and stores movement patterns. Skill acquisition progresses through three stages:

Stage	Characteristics	Coaching Approach
Cognitive (Beginner)	Conscious thought about every movement; inconsistent, error-prone; high mental effort	Simple clear cues; frequent demonstration; one focus point at a time; repetition on easy terrain

Stage	Characteristics	Coaching Approach
Associative (Intermediate)	Movements more consistent; self-detected/corrected errors; cognitive load decreases	Introduce complexity; video analysis; self-assessment after each attempt; vary conditions
Autonomous (Advanced)	Most movements automatic; can focus on tactics/route reading	Target specific weaknesses; practice under fatigue; high-variability training

VARIABILITY OF PRACTICE

Research shows practising a skill across varied conditions (hold types, angles, movement styles) produces more durable, transferable learning than repeating the same sequence. Resist having athletes repeatedly lap the same routes — variety is the engine of technical development.

Section 11: Cardiovascular & Respiratory Systems

The cardiovascular and respiratory systems deliver oxygen to working muscles and remove waste. While climbing isn't a cardiovascular sport traditionally, these systems determine recovery speed between attempts and sustained effort duration.

11.1 The Cardiovascular System

The cardiovascular system (heart, blood vessels, blood) transports oxygen and nutrients while removing CO_2 and waste. The heart is a four-chambered pump: the right side pumps deoxygenated blood to the lungs, the left pumps oxygenated blood to the body. Key variables:

- **Heart rate (HR)** — beats per minute; increases with intensity. Trained athletes have lower resting HR due to increased stroke volume.
- **Stroke volume (SV)** — blood ejected per beat; increases with aerobic training.
- **Cardiac output (Q)** — HR × SV; rises from ~5 L/min at rest to 20–25 L/min at maximal exercise in trained athletes.

Relevant to climbing coaches for three reasons:

1. **Aerobic fitness determines recovery speed** between attempts and rest holds.
2. **The occlusion effect** — sustained isometric grip restricts forearm blood flow, accelerating the 'pump'. ARC training develops capillary density to improve this.
3. **Monitoring training intensity** — heart rate helps ensure aerobic sessions stay genuinely aerobic.

11.2 The Respiratory System

The respiratory system exchanges gases between the body and environment via: nasal passages/mouth → trachea → bronchi → bronchioles → alveoli. Resting breathing rate (12–15/min) can rise to 40–60 during intense effort. Climbing-specific coaching points:

- **Breath-holding on crux moves** — elevates CO_2, increases perceived effort, reduces fine motor control. Cue rhythmic breathing.
- **Exhalation on effort** — reduces intra-abdominal pressure, supports core stability, keeps the athlete calm. Opposite of the unsustainable Valsalva manoeuvre.
- **Breathing at rest positions** — fully exhaling and resetting breathing at every rest directly extends climbing duration.

COACHING CUE — BREATHING

Teach climbers to exhale on the most demanding part of each move, and make a deliberate 'breathing check' at every rest hold. Three controlled breaths at each rest dramatically extends performance on sustained routes.

Section 12: The Three Energy Systems

All physical activity requires ATP. The body regenerates it via three metabolic pathways, all operating simultaneously — what changes is which system contributes most at a given moment.

12.1 The ATP-PC System (Phosphocreatine System)

The fastest, most immediate energy pathway, using stored phosphocreatine (PCr) to regenerate ATP. No oxygen required, no lactic acid produced — but very limited in duration.

- **Duration:** 0–10 seconds at maximal effort
- **Recovery:** PCr fully replenished within 3–5 minutes of complete rest
- **Climbing applications:** A single hard dyno, one-move deadpoint, short campus move, opening power moves of a boulder problem

TRAINING IMPLICATION — REST FULLY

To train the ATP-PC system effectively, athletes must rest fully between attempts — typically 3–5 minutes minimum. Resting only 1–2 minutes means the glycolytic system increasingly takes over, producing a different, less desirable training stimulus.

12.2 The Glycolytic System (Anaerobic / Lactic System)

Beyond 10 seconds of high-intensity effort, the body relies on glycolysis — breaking down glucose without oxygen, producing pyruvate. When glycolysis outpaces the aerobic system, hydrogen ions (H+) accumulate.

- **Duration:** 10–90 seconds at high intensity
- **By-products:** Hydrogen ions (H+) — lower muscle pH, cause the burning 'pump' sensation
- **Recovery:** 5–10 minutes significant; 20–30 minutes more complete lactate clearance
- **Climbing applications:** Sustained hard boulder problems, pumpy sport crux, 4x4 circuits, power endurance training

COMMON MISCONCEPTION

Lactic acid itself does not cause the burning sensation or failure — it's the hydrogen ion (H+) accumulation lowering pH. Lactate is actually a useful fuel source. Better-trained athletes clear hydrogen ions more efficiently — what 'lactate threshold training' actually develops.

12.3 The Oxidative System (Aerobic System)

For sustained, lower-intensity activity, the body uses oxygen to metabolise carbohydrates, fats, and some protein. Produces far more ATP per glucose molecule than glycolysis, but responds more slowly.

- **Duration:** 90 seconds and beyond; indefinitely with adequate fuel and oxygen

- **By-products:** Water and CO_2 — easily removed
- **Climbing applications:** Long moderate routes, approach hiking, ARC training, warm-up/cool-down laps, active recovery

12.4 Energy System Summary and Climbing Application

Energy System	Climbing Activity	Best Training Method
ATP-PC (0–10 sec, maximal)	Single hard dyno, campus move, maximal crimp	Max hangs (7–10 sec); 3–5 min full rest between attempts
Glycolytic (10–90 sec, high)	Sustained hard bouldering, pumpy sport crux, power endurance	4x4 circuits, linked boulder problems; 5–10 min recovery between sets
Oxidative (90 sec+, low-moderate)	Long moderate routes, ARC training, active recovery	ARC traversing (20–45 min), long easy routes

In practice, no session uses a single energy system in isolation. A typical session blends ATP-PC demands on hard problems, glycolytic demands during linked attempts, and oxidative demands during warm-up and easy traversing. Effective programming progressively develops all three.

Summary of Module 01

A quick reference before moving on

Key Terms & Definitions

Anatomical position	The standard reference position: standing upright, arms at sides, palms forward, feet slightly apart, toes forward.
Axial skeleton	Skull, vertebral column, and bony thorax. Protects the CNS and vital organs.
Appendicular skeleton	Bones of the limbs and shoulder/pelvic girdles. Enables movement through joints.
Synovial joint	The most mobile joint type, with a fluid-filled capsule. E.g. shoulder, elbow, hip, knee.
Tendon	Connects muscle to bone; transmits contraction force. Poor blood supply — adapts slowly.
Ligament	Connects bone to bone; provides joint stability and limits excessive ROM.
A2 pulley	The annular pulley on the proximal phalanx; most commonly injured structure in climbing.
Agonist	The prime mover in a movement — provides the major force.
Antagonist	Opposes the agonist; under-development is a primary cause of climbing injury.
Concentric contraction	Muscle shortens under load — the overcoming phase (e.g. pulling up).
Eccentric contraction	Muscle lengthens under load — the resisting phase. Primary stimulus for tendon adaptation.
Isometric contraction	Muscle length unchanged — force with no movement. Dominant type in climbing.
Sliding filament theory	Myosin heads bind actin and pull toward the sarcomere centre; driven by ATP, regulated by calcium.
Motor unit	A motor neuron and all fibres it innervates. All fibres contract simultaneously.

Stretch reflex	A spinal cord reflex triggered by rapid stretch; foundation of the SSC.
Stretch-shortening cycle	Pre-stretching before contraction generates more force via spindle loading, elastic energy, optimal length.
ATP	Adenosine triphosphate — the body's universal energy currency.
Phosphocreatine (PCr)	Regenerates ATP during maximal short efforts (0–10 sec). Replenished in 3–5 min rest.
Occlusion effect	Blood flow restriction during isometric contraction — primary driver of forearm pump.
ARC training	Aerobic Restoration and Capillarisation — sustained low-intensity climbing developing capillary density.
Proprioception	The body's sense of position/movement without visual input. Critical for footwork.
Neural fatigue	CNS fatigue after maximal training; requires 48–72 hours recovery.
Motor learning	Nervous system acquiring movement patterns through practice — cognitive, associative, autonomous stages.
Type I fibres	Slow-twitch, fatigue-resistant, aerobically fuelled fibres.
Type IIb fibres	Fast-twitch power fibres — high force, fatigue quickly, anaerobic.

Key Takeaways

- The musculoskeletal system — bones provide structure, synovial joints permit movement, tendons transmit force. The A2 pulley is the most injury-prone structure in climbing.
- Anatomical terminology and planes of movement — a shared language for precise communication. Climbing is multi-planar.
- Muscle roles — agonists, antagonists, synergists, and fixators act simultaneously. Antagonist undertraining is the leading cause of overuse injury.
- Muscle contraction types — concentric, eccentric, isometric. Climbing is predominantly isometric, explaining forearm pump.
- Muscle fibre types — Type I, IIa, IIb each suit different training goals and athlete weaknesses.
- Contraction physiology — the sliding filament theory, motor unit recruitment, rate coding, and the SSC underpin force generation.
- Cardiovascular and respiratory systems deliver oxygen and remove waste, determining recovery speed between attempts.
- The nervous system controls movement through motor units and proprioception. Neural fatigue needs 48–72 hours recovery.
- The three energy systems — ATP-PC, Glycolytic, and Oxidative — all contribute to every climbing session.

End of Lesson 03 and Module 1. Return to corelearning.co.za to complete the Lesson 03 quiz (80% pass mark) and unlock Module 02.